EXPERIMENT 01

```
clc;
clear;
close all;
Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20*fc
t = 0:1/fs:4/fm;
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
mu = Am/Ac;
am = Ac*(1 + mu*sin(2*pi*fm*t)) ...
    .* sin(2*pi*fc*t);
% Demodulation
dem = abs(am);
dem = movmean(dem, 20);
dem = dem - mean(dem);
% Display all graphs
figure('visible','on');

subplot(2,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
```

```
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,3);
plot(t,am);
title('AM Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,4);
plot(t,dem);
title('Demodulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
drawnow;
```

